Localizing Structure in Individual Differences: A Visual Illusion Case Study

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Data, analysis scripts, and code used to generate the figures are available at: https://github.com/specl/ind-illusions

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Abstract

Are people who are susceptible to one illusion susceptible to others? Previous research has shown small correlations, but might small values reflect attenuation from measurement error from trial-to-trial variation? To assess measurement error, we develop a set of novel data visualizations and hierarchical models. Data from 149 participants on 2 variants of 5 illusions were collected using an adjustment paradigm. The results showed low trial-noise and strong between-subject variability (e.g., signal-to-noise ratio ≈ 1.14, reliability ≈ 0.93). Correlations across illusions are low, around 0.22 ± 0.07. A Bayesian hierarchical analysis reveals minimal attenuation from measurement error in these values. Though correlations are low, latent variable analysis reveals a common latent factor that loads on all tasks and explains about 23.3% of the variance in illusion susceptibility.

Keywords: Individual Differences, Visual Illusions, Factor Analysis, Hierarchical Models, Reliability

Localizing Structure in Individual Differences: A Visual Illusion Case Study

Perceptual illusions have long been a topic of fascination to both the general public and scientists alike. The reason for this interest is straightforward: through the course of daily life, visual perception is primarily veridical and objects typically appear as they are. Yet specific configurations can lead to surprising and counter-intuitive distortion of our perception. An example of such a configuration is the Brentano illusion, depicted in the first row of Figure 1. Although the blue and the red lines are the same length, the surrounding arrow heads leads to the illusion than one of the colored lines is longer than the other.

For scientists, visual illusions offer a window into the processes and mechanisms underlying perception (Coren & Girgus, 2020). Recently, individual differences have been focal (see Tulver, 2019). Are people who are particularly susceptible to one illusion also more or less susceptible to another illusions? The hope is that the pattern of correlations that are diagnostic of latent processes or representations. Specific patterns in correlations may indicate factors or clusters. For instance, perhaps size-constancy illusions cluster together but are independent of illusions based on the misattribution of 3-D structure.

There is an active literature on individual differences in susceptibility to various illusions. Most studies report no common mechanism underlying perceptual illusions (Bosten et al., 2017; Cottier et al., 2023; Cretenoud et al., 2019; Grzeczkowski et al., 2017; Mazuz et al., 2023; Ward et al., 2017). These studies often find significant correlations within variants of specific illusions, yet they find only weak correlations between different types of illusions (with few exceptions). The resulting conclusion is that factors influencing susceptibility to illusions are likely multiple and specific rather than singular or general. These results concord well with a lack of general factors in visual perception studies, where a prevailing explanation is that individual variation likely reflects uncorrelated biological aspects such as retinal structure (e.g., Bosten & Mollon, 2010; Mollon et al., 2017). While this lack of common process characterizes much of the literature, there is a notable exception. Makowski et al. (2023) finds a single factor that operates across a wide class of illusions. These authors claim this general factor is positively related to positive personality traits (such aagreeableness and honesty-humility) and negatively related to maladaptive personality traits (such as antagonism, psychoticism, disinhibition, and negative affect).

There are several methodological difficulties in studying patterns of correlations from individual differences (Rouder et al., 2023). These difficulties arise from a statistical issue shown in Figure 2. The left-hand panel shows true scores across a group of participants in two tasks. The right-hand panel shows observed scores, and these are perturbed by trial noise. The result of this perturbation is that observed correlations are lower than true values. This downward bias, called the attenuation of correlation, is well known (Spearman, 1904). Even so, it is starkly different than most biases we are accustomed to in experimental psychology. In much of experimental psychology, we have a large-participant safety net. If we add more participants, then our estimates become closer to the true value. This limit does not hold for attenuation of correlation, however. For the case in Figure 2, adding more participants adds more points in the scatter plot, but it does not change the size of the perturbations from trial noise nor does mitigate the attenuation. In fact, adding more participants results in more confidence in a bad result.

Researchers occasionally ignore this fact or are, perhaps, unaware of its impact. They do so by stressing the number of participants in a design without sufficient consideration to trial noise and its problematic effects. It is routine for researchers to emphasize the number of participants in abstracts and methods without mentioning trial size or trial noise. It is also routine for researchers make power calculations and publish confidence intervals that reflect participant variability alone without consideration of attenuation. All these practices are flawed when there is much trial noise— they all stress being confident in a wrong answer.

What to do? Matzke et al. (2017) were perhaps the first to propose modeling trial noise along with variability across people and their correlations across tasks in a hierarchical framework. Rouder and colleagues (Rouder & Haaf, 2019; Rouder & Mehrvarz, 2024; Rouder et al., 2023) show that correlations are indeed disattenuated in hierarchical models, and Mehrvarz and Rouder (2025) show that accurate estimation of these correlations is possible but only under certain conditions and with certain assumptions. One of the key messages in this work is that some of the usual concepts in individual-difference research, such as reliability, are outdated because they do not take into account trial noise. And as such, there is a need for new ways of visualizing and communicating results for individual differences.

We have two goals. First, we study the nature of individual differences in visual illusions to uncover an underlying factor structure. Second, we use individual-differences in visual illusions to show how trial noise can and should be incorporated in research. We provide relatively new measures and ways of plotting data as well as new hierarchical factor models for inference.

To study individual differences in illusions, we constructed a battery of tasks. In each task, the participant had to adjust an object so that it matched another part of the display. For example, in the Brentano Illusion, participants adjusted the middle chevron until the size of the blue and red segments appeared equal in length. We build a pipeline in the analysis section. To preview, much effort is spent graphically examining how well we could measure each person's bias or score on each task. It is only after demonstrating that people differ and that we could reliably measure these differences that we perform a latent-variable decomposition of the correlations among the tasks. To foreshadow, we find a middle position inline with most of the previous research. First, although correlations across illusions are modest, their magnitude is not due to measurement error or attenuation. Individual differences in visual illusions are large, and we can localize correlations with high precision. Second, these modest correlations form a reliable positive manifold, revealing a common factor—susceptibility to illusions—which accounts for approximately 23.3% of the variance. The remaining variability appears largely idiosyncratic to specific illusions.

Methods

A demonstration version with all the tasks may be found at the following link[1].

Ethics

All methods employed in this study adhered to relevant guidelines and regulations, such as the Declaration of Helsinki. The Institutional Review Board (IRB) at the University of California, Irvine (UCI) approved the study protocol and experimental procedures under IRB approval number #3292. Prior to participating in the study, participants provided written informed consent and were made aware that their participation was entirely voluntary and that they could withdraw at any time.

Planning

There are two sample sizes to be set in an experiment. The first is the number of trials per person per task; the second is the number of people. These sample sizes play different roles. The first, the number of trials, is paramount as it determines the systematic attenuation and the expected correctness of any subsequent estimate of correlation. The second, the number of people, is about precision, and it makes little sense to have high precision in a bad estimate.

Rouder and Mehrvarz (2024) provide a means of planning the number of trials needed in a task. We used their approach, along with the visual illusion pilots they published, to plan our number of trials. Given the degree of trial noise, 15 trials per person per task was sufficient for a test-retest reliability of over .9. The number of participants reflects the precision needed to adjudicate between the two main hypotheses that there are no or small correlations across tasks vs. there is a common, one-factor solution. We settled on 150 individuals because it resulted in correlation estimates that had a worst-case precision of $\pm.06$. We do not use significance tests here, but if one did, the following power

[1] https://specl.socsci.uci.edu/acquire/data4/iBat5/demo.html

calculation might be informative. Suppose we wished to discriminate single task-wise correlations of.25 in value from chance with a power of .8 with a one-tailed test (negative correlations are not interpretable and should be taken as evidence for null in significance testing). This test is well powered at 98 individuals; hence the chosen value of 150 people is reasonable.

Participants

One-hundred and fifty individuals, recruited through Prolific, served as participants in the experiment. The only constraint was that participants were located in the United States. Participants were paid $5 USD for the 20 minutes it took to complete the battery.

Demographics

After the data-cleaning process, 138 participants remained. The average age was 35.34 years (SD = 11.01). There were 66 females, 71 males, and 1 participant who preferred not to say.

Tasks

The task battery comprises five visual illusions: the Brentano Illusion (Brentano, 1892), the Ebbinghaus Illusion (Ebbinghaus, 1913), the Poggendorff Illusion (Day & Dickinson, 1976), the Ponzo Illusion (Ponzo, 1910), and the Zöllner Illusion (Judd & Courten, 1905). These illusions are shown in Figure 1. In each task, participants adjusted a critical element of the stimulus. In the Brentano illusion (Panels A), participants adjusted the middle chevron to make the blue and red lines appear to be of equal length, with a tendency to set the red line too long and the blue line too short. In the Ebbinghaus Illusion (Panel C), participants adjusted one of the inner circles to match the other in size. In the Poggendorff Illusion (Panel D), participants aligned one of the two disjointed segments (left and right levers) to form a straight line. In the Ponzo Illusion (Panel E), participants adjusted one of the red lines to be the same height as the other red line. Lastly, in the Zöllner Illusion (Panel F), participants adjusted the horizontal lines to appear parallel. The method of adjustment was employed as it is long-standing (Fechner,

1860) and effective in measuring the strength of illusions (Cretenoud et al., 2021).

For each task, we constructed two versions. The two versions of the Brentano Illusion are shown in Panels A and B. The motivation for using two versions is to control for idiosyncratic response biases. If a participant ran only the version in Panel A and had a natural tendency to set the left segment too large at the expense of the right segment, this tendency would manifest as a larger illusion effect. To control for such a possibility, no matter how remote, we constructed an alternate version so that such a tendency would not greatly impact the true illusion effect. Alternate versions of each illusion were constructed analogously. For the Ebbinghaus Illusion, in Version 1, participants adjusted the right inner circle (surrounded by smaller circles); in Version 2, they adjusted the left circle (surrounded by larger circles). For the Poggendorff Illusion, in Version 1, participants adjusted the right lever vertically. In Version 2, the illusion was flipped vertically, with the downward-facing right lever being adjusted on the right-hand side. For the Ponzo Illusion, in Version 1, participants adjusted the closer red line, and in Version 2, they adjusted the farther red line. Lastly, Zöllner Version 1 is depicted in Panel F of Figure **??**, and Version 2 was the same but mirrored horizontally. We expected the magnitude of an illusion for an individual to be consistent across versions. This expectation held for some illusions, but surprisingly, not for others. As such, we could not collapse across the versions and they feature prominently in the subsequent analyses.

Design

The design was a 5×2 within-subject factorial design. The first factor was the task (the specific illusion), and the second factor was the task version. We chose to repeat each task-by-version combination 15 times for each individual. This trial size of 15 was chosen as follows. We ran a pilot experiment to assess the size of illusions, the variability between people, and the variation across replicate trials. We then used the method described in Rouder and Mehrvarz (2024) to obtain an expected reliability of over .95 per illusion. The method yielded 30 trials for each illusion which we split across versions. The response measure was how far the adjustment varied from the ground truth.

Procedure

Participants registered through Prolific, and were then referred to our webserver for the experiment. After reading the consent letter, they were asked to raise the volume on their speakers so they could hear feedback tones and to size their screen so that they could see all stimulus elements. Participants ran two cycles, and in each cycle, each of the five illusions was presented in a random order. Versions were randomly chosen in the first cycle, and the other version was presented in the second cycle. For each illusion in the cycle, a set of instructions and two practice trials were provided, and then the participant ran 15 replicate trials. Across all tasks and cycles, there were 150 trials in total. Participants had to adjust elements as discussed previously using the arrow keys on their keyboard, and each press moved the critical element a small amount. When they were satisfied with their adjustment, they pressed the spacebar, which advanced to the next trial. Breaks were given after tasks of 15 trials.

We found in a pilot that participants varied greatly in how much time they took before being satisfied with their adjustment. Some would spend tens of seconds going back and forth until they made the spacebar commitment. To keep the experiment fairly regular across individuals, it was necessary to impose an eight-second time limit. Participants could advance to the next trial by pressing the spacebar or wait the eight seconds for automatic advancement.

For each trial in each illusion, we varied a number of parameters. For convenience, several illusions have a *target* size, such as the size of the to-be-matched line segment (Brentano, Ponzo), circumference (Ebbinghaus), or position (Poggendorff). We jittered these target sizes as well as the starting sizes of the adjusted elements. We also jittered other elements, including the overall position on the screen and, in the Zöllner illusion, the length of the main lines and cross lines. This variation prevented the development of stereotypical responses across repeated trials.

Results

Dependent Measure

On each trial, participants produce an adjustment that may differ from the ground truth. We scored the illusion effect as follows: For the Brentano Illusion, the score was the number of pixels to the left (or right) of half of the horizontal segment relative to the size (in pixels) of the segment. Relative effects were used for the Ebbinghaus and Ponzo illusions, each effect was the difference between the adjusted value and the true value relative to the size of the true value. For the Poggendorff and Zöllner illusions, however, there is nothing to compare the adjustments against. Therefore, the scores were measured as the number of displaced pixels (Poggendorff) from the true value or the angle from horizontal (Zöllner). We call these values the "adjustment score" in the following and they are the relative bias in adjustment.

Data Cleaning

Data obtained online can be messy. Surprisingly, the data in this report were fairly clean, and we think this is because the battery was brief (it took under 20 minutes to complete) and participants reported that it was engaging. Because we had little expectations on the ranges of data *a priori*, we performed data cleaning after data collection but before analysis of pattern of correlations. A very detailed report on data cleaning is available at linked document[2]; here is a summary: Data-cleaning decisions fall into three camps: those made without any reference to the adjustment score, those made with reference to adjustment scores in each task in isolation, and those made with reference to adjustment scores across two tasks at a time.

Data Cleaning Without Reference To Adjustment Sores. We first made a set of cleaning decisions before any analysis of the adjustment score: I. One participant performed the task twice rather than once. The second session was excluded. II. We

[2] https://github.com/specl/ind-illusions/blob/public/_data-processing/data-cleaning.pdf

worried about participants who sped through the task at atypical and unreasonably fast speed. As a bit of context, the adjustment paradigm here consisted of making several keypresses to finely adjust objects. This process takes longer than conventional response times, and typical responses are in the 5-second range. The histograms of response times was bimodal with a separate mode below 600 ms. This mode was not large, and overall, only 1.5% of responses fell here. These responses were excluded, as were individuals who had more than 5% of their responses in this mode (5 individuals).

Data Cleaning Each Task In Isolation. We worried about participants who may not have understood some of the adjustment tasks. We conducted a univariate analysis to detect outliers based on standardized scores across different tasks and their respective versions. Participants whose z-score on any task exceeded 4 were removed (6 individuals). When these z-scores exceeded 4, they often did so dramatically indicating that these 6 individuals did not understand the instructions for one or more of the tasks.

Data Cleaning Across Bivariate Configurations. We worried that correlations across tasks might be unduly influenced by high-leverage points. We ran a suite of leverage tests (Difference in Fits, Difference in Beta Coefficients, and Cook's Distance (Belsley et al., 1980)). Although there were a few high leverage points, they were not unreasonable in any other way and we could find no rationale for excluding them.

Demographics

After the data-cleaning process, 138 participants remained. The average age was 35.34 years (SD = 11.01). There were 66 females, 71 males, and 1 participant who preferred not to say.

Tasks In Isolation

The goal of the analysis is to understand the relationships among the tasks. However, tasks must be suitable in that they measure a signal without an undue degree of trial noise. If individual task scores are too noisy, then correlations are attenuated too low, and latent-variable decompositions may be unstable (Karr et al., 2018; Rouder et al., 2023). The first step, then, is to provide a deep dive into the trial noise in the task scores themselves.

Size of the effects. When individual-difference studies are conducted, the focus is on covariation and correlation, and the size of the effects does not enter into the critical parts of the analyses. Yet, the effect sizes may serve as an indicator of the robustness of subsequent individual-difference analyses. When effects are large, there is substantial headroom for people to vary. When effects are small, there is far less headroom. A good first check, then, is the size of the effect

We computed an effect size and a t-value for every individual in each of the 10 tasks. The effect size for the ith individual in the jth task is $d_{ij} = m_{ij}/s_{ij}$, where m_{ij} and s_{ij} are respectively the sample mean and sample standard deviation of adjustment scores for all trials for the ith individual in the jth task. The t-value is $t_{ij} = d_{ij} \times \sqrt{K_{ij}}$. Figure 3A shows a histogram of all effect sizes and t-values, and they are nothing short of spectacular in value. Even though there are at most 15 trials per individual per task version, the vast majority of the effect sizes exceed Cohen's 0.8 criteria for large, and even more are significant at 0.05 level. Figure 3B shows these effects tabulated by task as empirical cumulative distribution functions. Even though there are differences across the tasks in mean effect sizes and the spread of effect sizes, the large effect sizes hold for all 10 tasks. In summary, visual illusions are so large that performance from just 15 trials contains more signal than typical cognitive studies comprised of thousands of trials.

Discriminating among individuals within a task. For the remainder of the analyses, we used scaled adjustment scores as follows. For each task, mean squared error (MSE) was computed among individuals, and the square-root is the standard deviation of task-specific residual variance. Then, each score for all individuals in a task was expressed in units of this standard deviation, that is, in task-specific effect size units. Let Y_{ijk} be the kth replicate observation for the ith person on the jth task. Let $s_j = \sqrt{\text{MSE}_j} = \left(\sum_{ik}(Y_{ijk} - m_{ij})^2/I(K-1)\right)^{1/2}$. Then the task-scaled score is Y_{ijk}/s_j.

Figure 4 shows these scaled scores in a configuration that we term *task plots*. Here, individuals' observed mean scaled scores in each task are ordered and plotted using dashed lines. The solid lines and shaded areas reflect regularized estimates (from a Bayesian hierarchical model discussed subsequently) and 95% credible intervals, and these intervals show how well each individual's score may be localized. There are substantial differences in scores across individuals in all tasks, indicating that performance can be discriminated to high degree across individuals. We recommend that task plots always be included as they provide a clear visualization of what is possible in terms of identifying and understanding individual differences.

Goodness of Task Measures. The usual way to understand the goodness of a task is reliability.[3]. There is, of course, a direct relationship. The larger the spread of the mean scores relative to CIs in the task plots in Figure 4, the higher the reliability. Yet, both task plots and the associated reliability measure strike us as incomplete for communication about tasks.

This incompleteness comes from the fact that reliability and task plots are directly influenced by the number of replicate trials. The Stroop task, for example, is unreliable with 15 trials, but it is very reliable with several thousand trials (Lee et al., 2023)[4]. What is needed for grounded communication is a task measure of signal-to-noise that does not depend on the number of replicate trials. We advocate the signal-to-noise ratio of a task as such a measure (Rouder & Mehrvarz, 2024), and denote it as γ^2. The signal, the numerator of γ^2, is the variability across individuals' true scores in a task. The noise, the denominator of γ^2, is the trial-to-trial variability.

The measure γ^2 may be thought of as a ratio version of ICC(1) (Liljequist et al.,

[3] One course is to split the trials, compute the correlation, and scale it up to the full sample size through the Brown-Spearman prophecy formula. This is a test-theory approach that fails to leverage the fact that experiments are comprised of multiple replicate trials. Reliability here is computed directly from within-trial and across-trial variabilities, see Rouder and Mehrvarz (2024).

[4] c.f. Burgoyne et al. (2023) for a high-reliability Stroop-like task where performance is averaged rather than contrasted across congruent and incongruent conditions.

2019), and we prefer it to ICC because we find the ratio-interpretation quite natural.

Tasks with high values of γ^2 are reliable even with few replicate trials per individual per task; tasks with low values are reliable only if there are a high number of replicate trials per individual per task. The γ^2 for each illusion is denoted in the task plot, and the average is 1.145. This value means that across illusions, variation across people is as big as variation within a trial. It is a large value by any standard, and it should be possible to understand individual differences to high precision.

Split-Half Reliability: A Sanity Check. In the preceding plots, we showed that the illusion effects are quite large and that individuals can be discriminated. We provide one final check on this sanguine state—a split-halves scatter plot. Plotted are individual mean effects for the second-half of trials as a function of the mean effects for the first-half of trials (Figure 5). As can be seen, the association—the split half reliability—is quite high.

In summary, with such large and reliable effects, it should be straightforward to assess the relations among tasks without excessive variability or attenuation.

Relations Among Tasks

Now that we have established that these tasks have very high signal-to-noise characteristics, the next step is analyzing the relations among them. Figure 6 shows the correlations among the tasks obtained in two different ways. The upper triangle shows the conventional sample correlations; the lower triangle shows estimates from a hierarchical Wishart model that disattenuates these correlations for trial noise. The hierarchical Wishart model is from Rouder et al. (2023), and it is comprised of two levels: a data level and an individual-effects level. At the data level, it is assumed that performance on a trial results is the sum of individual-by-task true score and trial noise. Let θ_{ij} denote this true score for the ith individual on the jth task, and let $\boldsymbol{\theta}_i = (\theta_{i1}, \ldots, \theta_{iJ})$ denote the vector of true task scores for the ith individual. At the individual-effect level, a multivariate random effects model is placed of $\boldsymbol{\theta}_i$: $\boldsymbol{\theta}_i \sim \text{Normal}_J(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The key here is that all relations among the tasks are in the variance $\boldsymbol{\Sigma}$. We estimate $\boldsymbol{\Sigma}$ in the Bayesian framework, and as such, a prior is needed. The Wishart prior (Wishart, 1928) serves as practical choice as it places little *a priori* constraint on $\mathbf{\Sigma}$ while ensuring that posterior values are valid variance matrix (that is, they are symmetric and positive-definite matrices). The advantage of the Bayesian framework is two-fold. First, it is conceptually straightforward approach for separating trial noise from across-individual variation. Second, analysis is computationally convenient with modern MCMC sampling methods (Gelfand & Smith, 1990). Posterior of correlations, derived form the variance matrix, are computed within each iteration of the MCMC chains. The posterior mean is shown in the lower triangle of Figure 6.

Comparisons of the sample correlations and model-based correlations reveal a tight correspondence indicating little attenuation in sample correlations. This lack of attenuation is relatively rare in experimental settings and reflects that effects are large relative to trial noise. First, the within-version correlations. For three illusions, the Ponzo, Poggendorf, and Zoellner, there are high correlations across the two versions of the illusions; for the other two illusions, the Brentano and Ebbinghaus, the correlations are more modest. This result is surprising. We had expected high correlations, and the task versions were included as a control for response biases. The modest correlations indicate that it is not wise to average scores across versions. Second, the cross-illusion correlations. Each of these is positive in direction and modest in value.

Uncertainty In Correlations. We recommend that researchers study the uncertainty in cross-task correlations. In designs where participants provide repeated trials per task score, there are two sources of uncertainty. One is from the noise across trials. The degree of uncertainty from this source reflects not only the degree of trial-to-trial variation, but also the number of trials in each cell. The more trials, the less influence this variation has on the uncertainty of the correlation coefficient. The other source reflects variation across individuals, and the more individuals the less uncertainty from this source. These sources may be combined into an overall assessment with the hierarchical Wishart model. The 95% posterior credible interval (CIs) serves as a suitable measure of uncertainty.

Figure 7 shows the mean and uncertainties (95% CIs) for the correlations. Consideration of them provides the following insights: First, as noted, many of them are undoubtedly positive providing evidence for a positive manifold across illusions. Accordingly, there is at least one factor in common. Second, the three highest correlations, which stand apart, are correlations among different versions of the tasks. If we focus on correlations across different illusions, there is not so much variability. These are clustered around .22 in value with credible intervals that have a full width of .134 ($\pm$.067) on average. This small value, .22, is by-and-large confirmation of previous results. One visual illusion accounts for about 4.7% of the variance in another illusion.

Latent Variable Analyses

Interpreting sample correlation such as those in Figure 6 is often aided by latent-variable models. Latent variable modeling, unfortunately, is a bit of a dark art in our view. While implementation and analysis is relatively straightforward in modern packages, making judicious choices in specification and providing sound interpretation is challenging (Armstrong, 1967; Preacher & MacCallum, 2003). In visual perception, Tulver (2019) recently notes that many of these issues, what is a factor, how to measure factors, and how to interpret factors, remain unresolved in the field. Here, even though the data are of high quality, we run head-long into these difficulties.

Principle Components. Many researchers who search for factor structure use principle component analysis (PCA). Table 1 shows the eigenvalues accounted for by each component. Accordingly, there are four factors with eigenvalues greater than 1.0.

PCA is often seen as interchangeable with factor analysis. Though similar, there are critical differences. For experimental psychological research, the latter is superior to the former. To see the key difference, consider the test case where there is no structure—that is, the correlation matrix is a diagonal matrix plus a small random variation from trial noise. The PCA for J tasks would yield J factors each accounting for about $1/J$ proportion of the variance. The factor model underlying factor analysis, however, has error terms that account for uncorrelated noise. For the test case, the residuals from these error terms would be large, and the factor loadings on illusions that account for the structure would be quite small. This difference has been well known (Velicer & Jackson, 1990) and goes under the moniker that factor analysis partitions common variance rather than total variance (Kline, 2023). Because factor models are much easier to interpret in low-signal scenarios, they should be used instead of PCA in vision, cognitive control, and related fields.

Classic Exploratory Factor Analysis. We performed exploratory factor analysis (EFA) separately for 1 to 6 factors. Table 1 also shows the model selection statistics, and as can be seen, the situation is ambiguous. While inspection of eigenvalues (first column) indicates support for a one-factor or four-factor solution, AIC indicates a five-factor solution and BIC indicates a three-factor solution. Examination of the varimax factor loadings for 5 factors is shown in Table 2. Factors tend to load on a single task with only modest cross-task loading. With this result, as well as the ambiguity of dimensions, it is difficult to gain much additional insight above that from the sample correlations.

Classic Confirmatory Factor Analysis. We also performed a series of confirmatory factor analyses with models shown in Figure 8. In the models in Figure 8A-B, the observed illusion scores load onto a latent variable. Correlations across the five latent variables are then computed (Fig 8A) or modeled with additional latent factors (Fig 8B). The main problem we encountered was that the resulting parameters had substantial negative variance components rendering them inadmissible. The problem is a known as a Heyward Condition (Rindskopf, 1984) and is under active consideration in the literature (Cooperman & Waller, 2022), We also tried a more flexible version of the model, Figure 8C), but encountered the same issues.

Bayesian Hierarchical Factor Models. We have recently developed hierarchical factor models (Rouder, Mehrvarz, & Stevenson, 2024). The first level of these models is similar to the hierarchical Wishart model discussed in the previous section. It accounts for trial noise through a linear framework that captures variability within individuals and tasks, allowing for the separation of trial noise from the true-score parameter θ_{ij}. The second level models these true-score parameters as coming from a latent factor structure:

$$\boldsymbol{\theta}_i \sim \mathcal{N}_J(\boldsymbol{\mu}, \boldsymbol{\Lambda}\boldsymbol{\Lambda}' + D(\boldsymbol{\sigma}^2)),$$

where $\boldsymbol{\mu}$ is the vector of task means, $\boldsymbol{\Lambda}$ is the factor loading matrix, and $D(\boldsymbol{\sigma}^2)$ is a diagonal matrix which task-specific unique variances. The crux of these models is the decomposition of the covariance into common and unique sources of variability across tasks (Preacher & MacCallum, 2003). This decomposition separates shared factor structures from task-specific residual variance. Additionally, factor analysis can take orthogonal or oblique forms depending on whether factor scores are assumed to be independent. The model described here assumes orthogonality, meaning that individuals who score high on one factor are, on average, no better or worse on another. In other words, the model treats abilities across different factors as statistically independent.

We conducted confirmatory factor-model analysis with seven factors [5] . The first five factors were illusion specific, e.g., a Brentano factor, and accounted for correlations across the versions of the task. The sixth and seventh factors were general in that they could load on several illusions. These two factors are the main target of study and they are estimated free of influence from across-version variation.

Table **??** presents the results of the model, and Figure **??** visualizes key components of the variance decomposition. The first five factors capture variance specific to differences between versions of each illusion. The key result is that sixth factor loads well on all 10 tasks. It explains more than four times the variance of the seventh factor. This pattern strongly supports the conclusion that a single general factor underlies susceptibility to visual illusions. A projection of each task-version on the two general factor is depicted by (Figure **??**A). However, it is important to note that unique variance between the two

[5] For full analysis and more detailed model specification, refer to Rouder, Mehrvarz, and Stevenson (2024).

versions of specific illusions is relevant (Figure **??**B), and unique variability from each version and task remains substantial (Figure **??**C). The general factor alone accounts for approximately 23.3% of the total variance—substantially higher than the 5% expected from squaring typical task-level correlations. In the discussion, we consider why these magnitudes differ and what they imply for interpreting the structure of individual differences in perceptual susceptibility.

Table **??** presents the results of the model, and Figure 9 visualizes key components of the variance decomposition. The first five factors capture variance specific to differences between versions of each illusion. The key result is that the sixth factor loads consistently across all ten tasks and explains over four times the variance of the seventh factor. This pattern strongly supports the conclusion that a single general factor underlies susceptibility to visual illusions. A projection of each task version onto the two general factors is shown in Figure 9A. However, unique variance between the two versions of each illusion remains relevant (Figure 9B), and unique variance attributable to specific versions of each illusion remains substantial (Figure 9C). The general factor alone accounts for approximately 23.3% of the total variance, which is substantially greater than the 5% expected from squaring typical task-version-level correlations. In the discussion, we consider why these differences occur and how to interpret them in drawing scientific conclusions.

Discussion

In this paper, we paid equal attention to the substantive issue of how visual illusions covary across individuals and the methodological issue of how to document structure in this covariation. The following conclusions may be drawn substantively about visual illusions:

1. Visual illusions are strikingly large in effect and straightforward to document. Data from each of our participants in each task, comprising at most 15 observations, often shows more evidence of an effect than typical cognitive psychology experiments comprised of hundreds of observations from scores of participants.

2. Individual differences are large. The amount of variance across individuals is

large relative to trial noise. The ratio, γ^2 is around 1.145-to-1 (in variances), which is about 73 times larger than those in cognitive control tasks (Rouder, Peña, et al., 2024).

3. Correlations across different illusions are relatively small. This small size is not due to statistical attenuation but reflects the true state of things—illusions are at most weakly related.

4. Nonetheless, there is at least one common factor, moreover, this factor accounts for 23.3% of the total variance.

Our methodology for uncovering the structure of individual differences has a few elements that are not commonly stressed:

1. We spend much effort assessing how much individual variation is recoverable. The task plots in Figure 4 show how different individuals are in each effect. If these differences are hard to discern graphically, then it may be hard to document correlations among tasks.

2. Understanding the uncertainty in correlations is key to placing them in context and performing subsequent latent-variable modeling. The plot in Figure 7 shows immediately that with the exclusion of version correlations, correlations are modest and fairly uniform, and moreover, there is a positive manifold across them. The modesty of these correlations relative to uncertainty means that it is not possible to derive a many-factor solution. One issue is that obtaining these uncertainties is not straightforward. They come from a hierarchical model where the data model accounts for trial-to-trial variation and the model of individuals accounts for between-participant covariation. Bayesian hierarchical models are perfect for this endeavor and there are a wide range of successful applications and tutorials for psychologists.

3. Latent-variable modeling was difficult on several accounts. The main problem was accounting for high version correlations for some illusions while accounting for lower version correlations in others. We relied on the inspection of the correlations for guidance.

Interpretation

In the literature, there are seemingly two incompatible results. The first is that individual differences in visual illusions are at best weakly correlated (Cretenoud et al., 2019) and the proportion of variance accounted for any illusion on any other is surprisingly small. The second is that individual differences are related by a common factor which may even reflect personality differences (Makowski et al., 2023). We find support for both positions. Visual illusions are undoubtedly weakly related. Individual differences in any one illusion account for 5% of the variance in another. And, visual illusions are connected by a common factor that accounts for 23.3% of the variance in each illusion when one version of each is entered into the analysis. Of course, both proportions of variance are correct—this is how latent variables work. One task may explain a_1 of the variation in the latent variable; the latent-variable may explain a_2 of the variation in another task, and, assuming these proportions are independent, then the first task explains $a_1 \times a_2$ of the variance in the second.

In the introduction of this paper, we raised the hope that patterns of individual differences could inform about the structure of perception and cognition that underlie these illusions. That hope remains unrealized. There most definitely is common variation, but it seems too small to finely decompose, and, consequently, too small to understand the perceptual and cognitive factors that mediate illusions. Moreover, because this variation is small and loads on all tasks, it may have little to do with illusions per se. An alternative is that the factor relates to how people engage in computer tasks or how conscientious they are (Makowski et al., 2023).

Future Directions

The set of results that strikes us the most as cognitive psychologists is comprised of version correlations. For some illusions, a small change in setup designed to reverse the direction of the illusion does exactly this. Consider the Zöllner illusion. In Figure 1F, the top horizontal line appears slanted such that the left end is higher than the right end. In the alternative version (not shown), the cross lines are rotated 90 degrees. The bias reverses as expected—the top horizontal line appears slanted such that the left end is lower than the right end. And, the correlation is high—people who show larger biases in one version show large reverse biases in the other. This reasonable expectation does not hold for the Brentano or the Ebbinghaus Illusions. For instance, in the Brentano illusion (Figure 1A), people set the chevron too far to the left. In the alternative version (Figure 1B), they set it too far to the right. These biases correlate moderately with other illusions, such as the Zöllner and Poggendorf. Yet, the two versions have surprisingly little correlation between themselves. Neither of these results is a statistical fluke as the data are highly reliable and the 95% credible intervals on the correlation coefficients are well separated (see Figure 7). How is this pattern so? What about the Brentano and Ebbinghaus Illusions make them pathological in this regard? Perhaps, rather than clustering illusions based on how well they correlate to each other, we should study how well they correlate within themselves, that is, how robust they are to reasonable configurable changes. It may be this robustness that provides insight into the mechanisms after all.

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Declarations

Ethics Approval. All procedures were approved by the Institutional Review Board of the University of California, Irvine (IRB Protocol #3292).

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Open Science Practices. The data, analysis scripts, and code used to generate the figures are openly available at https://github.com/specl/ind-illusions.

Conflicts of Interest. The authors declare no conflicts of interest.

Consent to Participate. Informed consent was obtained from all individual participants included in the study.

Author Contributions. Mahbod Mehrvarz and Jeffrey N. Rouder jointly conceptualized the study, wrote, and revised the manuscript. Mahbod Mehrvarz developed the models, conducted the analyses, and created the figures. All three authors contributed to data collection, preprocessing, and quality control. All authors approved the final manuscript.

Table 1

Exploratory Factor Analysis Model Selection Statistics.

Factors	PCA Eigenvalue	AIC	BIC	CFI	RMSEA
1	3.239	3734.880	3793.425	0.463	0.220
2	1.545	3630.132	3715.022	0.724	0.183
3	1.261	3554.424	**3662.733**	0.917	0.121
4	**1.183**	3537.297	3666.097	0.972	0.089
5	0.771	**3527.484**	3673.847	1.000	0.000
6	0.682	3536.200	3697.198	1.000	0.000

Note: Boldface typeset indicates number of selected factors. These are 4 by eigenvalues, 5 by AIC and 3 by BIC.

Table 2

Exploratory Factor Analysis Loadings

Task-Version	Factor Loadings					Proportion of Variance	
	1	2	3	4	5	Unique	Shared
Brentano v1	*0.401*	0.145	*0.279*	0.124	-0.119	0.711	0.289
Brentano v2	*0.315*	0.008	*0.336*	0.019	0.041	0.785	0.215
Ebbinghaus v1	0.096	0.014	0.124	0.223	**0.724**	0.401	0.599
Ebbinghaus v2	0.163	0.043	0.103	**0.854**	0.267	0.160	0.840
Poggendorf v1	0.119	0.072	**0.904**	0.104	0.013	0.152	0.848
Poggendorf v2	0.154	0.101	**0.704**	0.021	0.155	0.446	0.554
Ponzo v1	0.189	**0.811**	0.092	0.175	-0.115	0.254	0.746
Ponzo v2	0.072	**0.976**	0.094	-0.113	0.144	0.000	1.000
Zoellner v1	**0.870**	0.147	0.034	0.080	0.181	0.182	0.818
Zoellner v2	**0.771**	0.070	0.209	0.081	0.049	0.348	0.652

Note. Boldfaced loadings highlight the relations between factors and tasks.

Table 3

Factor Loadings For CFA Model of Visual Illusions.

Task	**Nuisance Factors**					**Exploratory Factors**		**Unique Variance**
	F1	F2	F3	F4	F5	F6	F7	
Brentano 1	0.270	–	–	–	–	0.569	-0.073	0.477
Brentano 2	-0.075	–	–	–	–	0.494	-0.007	0.539
Ebbinghaus 1	–	0.621	–	–	–	0.251	0.160	0.246
Ebbinghaus 2	–	0.625	–	–	–	0.354	-0.040	0.249
Poggendorf 1	–	–	0.623	–	–	0.528	-0.012	0.393
Poggendorf 2	–	–	0.565	–	–	0.508	0.066	0.342
Ponzo 1	–	–	–	0.824	–	0.352	-0.011	0.194
Ponzo 2	–	–	–	0.842	–	0.272	0.110	0.165
Zöllner 1	–	–	–	–	0.588	0.574	0.011	0.321
Zöllner 2	–	–	–	–	0.555	0.616	-0.042	0.298
Prop. Var.	0.021	0.081	0.078	0.141	0.069	0.233	0.055	0.322

Note. We fit a 7-factor hierarchical Bayesian model. The first five factors were used to capture within-version idiosyncrasies, with all other loadings zeroed out for other tasks. The last two factors were general and shared across all versions and tasks. All loadings were scaled to reflect variance decomposition in the correlation metric, such that each value represents a factor's contribution to shared variance. The first general factor explained over four times as much variance in the correlation structure as the second. For more details, refer to Rouder, Mehrvarz, and Stevenson (2024).

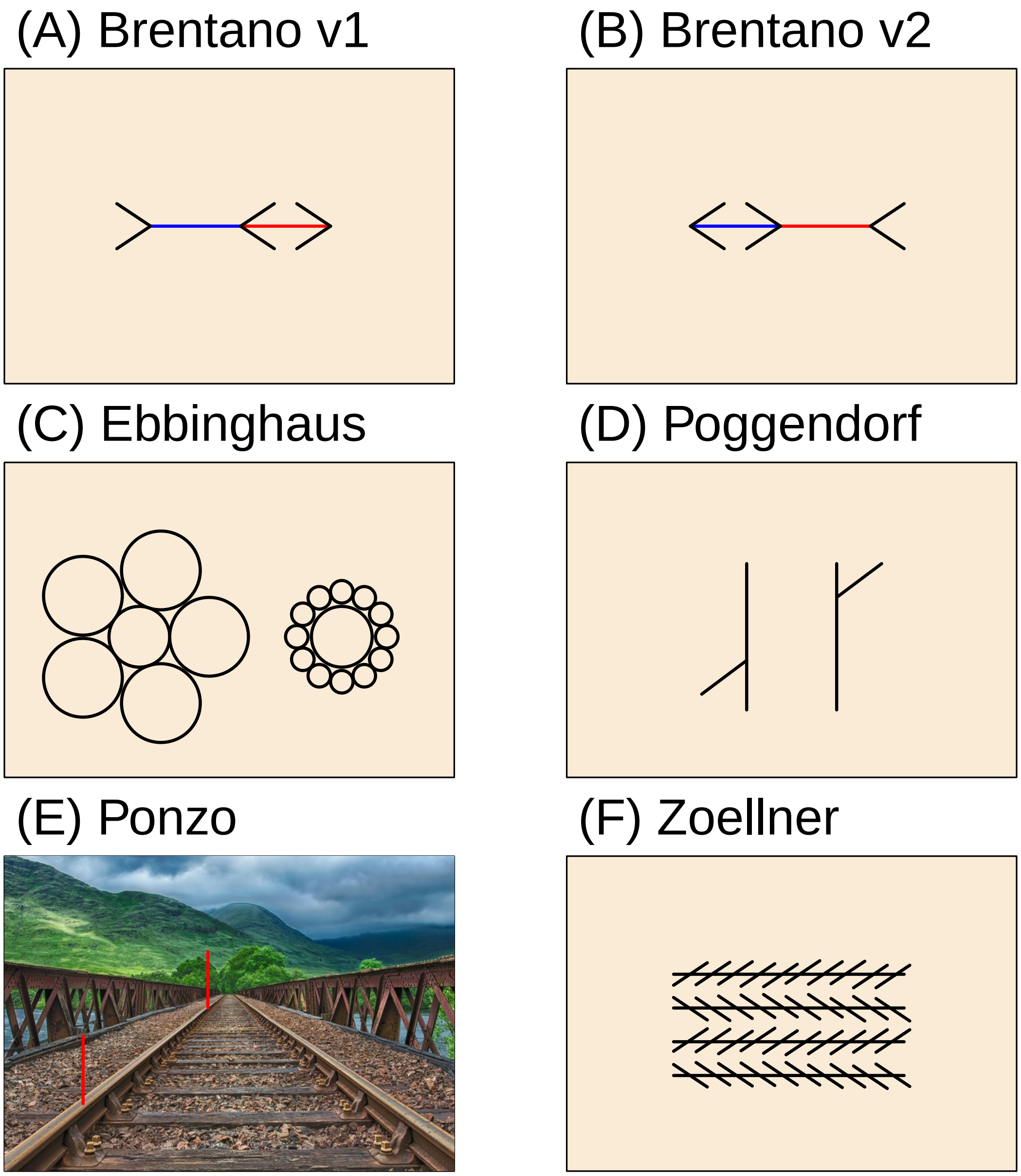

Figure 1

Illusion Tasks: *A-B. Two versions of the Brentano Illusion were presented. Participants adjusted the center chevron so that the red and blue lines appeared equal in length. C. In the Ebbinghaus Illusion, participants adjusted the right inner circle in Version 1, while in Version 2, they adjusted the left inner circle. D. For the Poggendorf Illusion, participants adjusted the right lever vertically in Version 1. In Version 2, the illusion was flipped vertically, with the downward-facing right lever being adjusted on the right-hand side. E. In the Ponzo Illusion, participants adjusted the closer red line in Version 1 and the farther red line in Version 2. F. For the Zöllner Illusion, participants adjusted the horizontal lines to appear parallel. Version 1 is depicted in the figure and Version 2 was exactly the same but mirrored horizontally.*

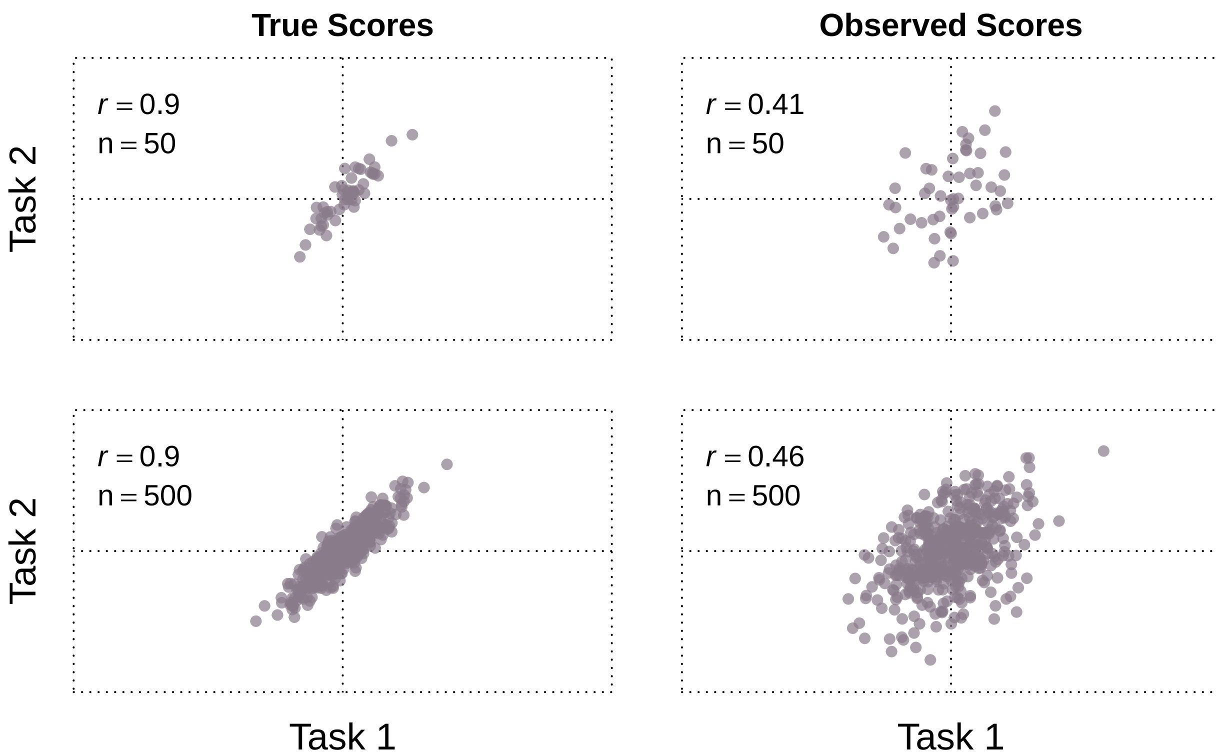

Figure 2

Trial noise leads to attenuation of correlations, with the left panel showing the true correlations between Task 1 and Task 2, and the right panel showing the observed correlations after introducing noise. Increasing the number of participants (n) does not mitigate this attenuation, as shown by the similar attenuation in correlation from true to observed scores regardless of the number of participants (n = 50 or n = 500).

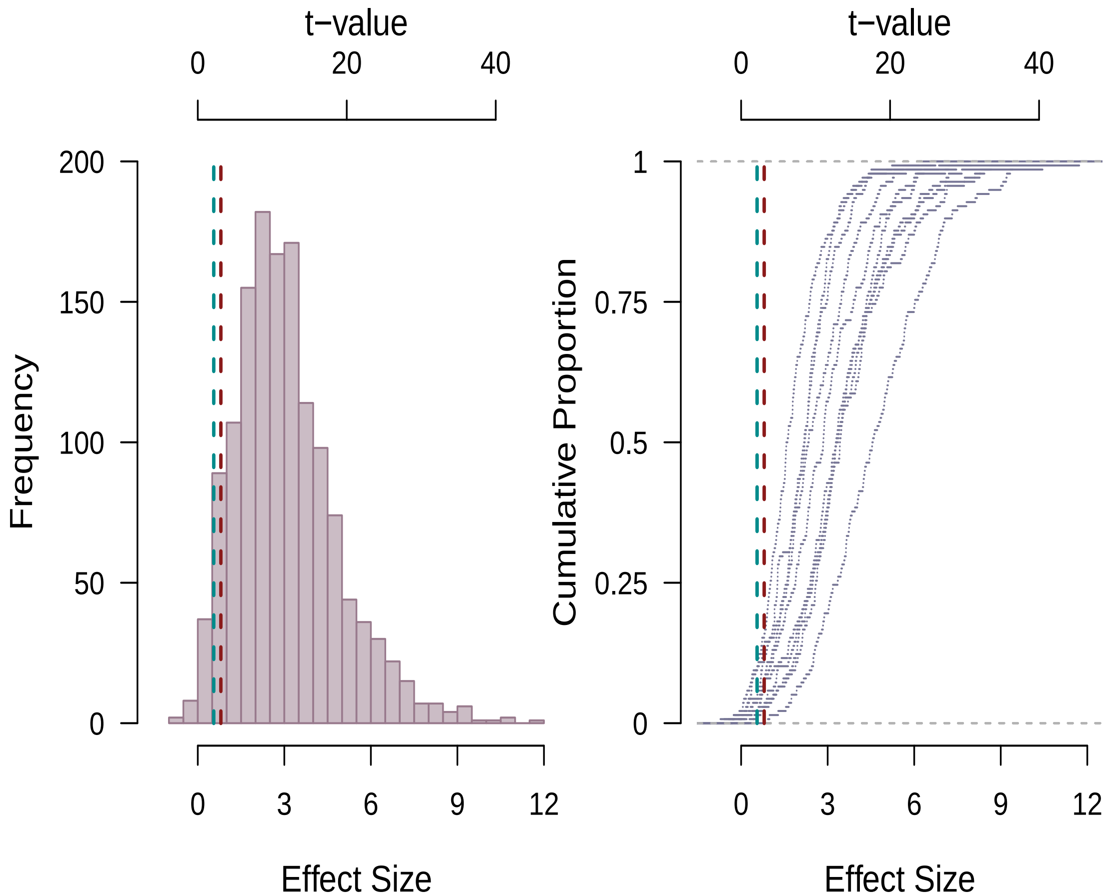

Figure 3

The size of illusions effects. The data from each participant in each task is treated as a separate experiment. Left: Histogram of all effect sizes shows large effects for almost all combinations of individuals and tasks. Right: The same data plotted as an empirical-cumulative-distribution function reveals although there are large effects for all tasks, there are some differences across the tasks.

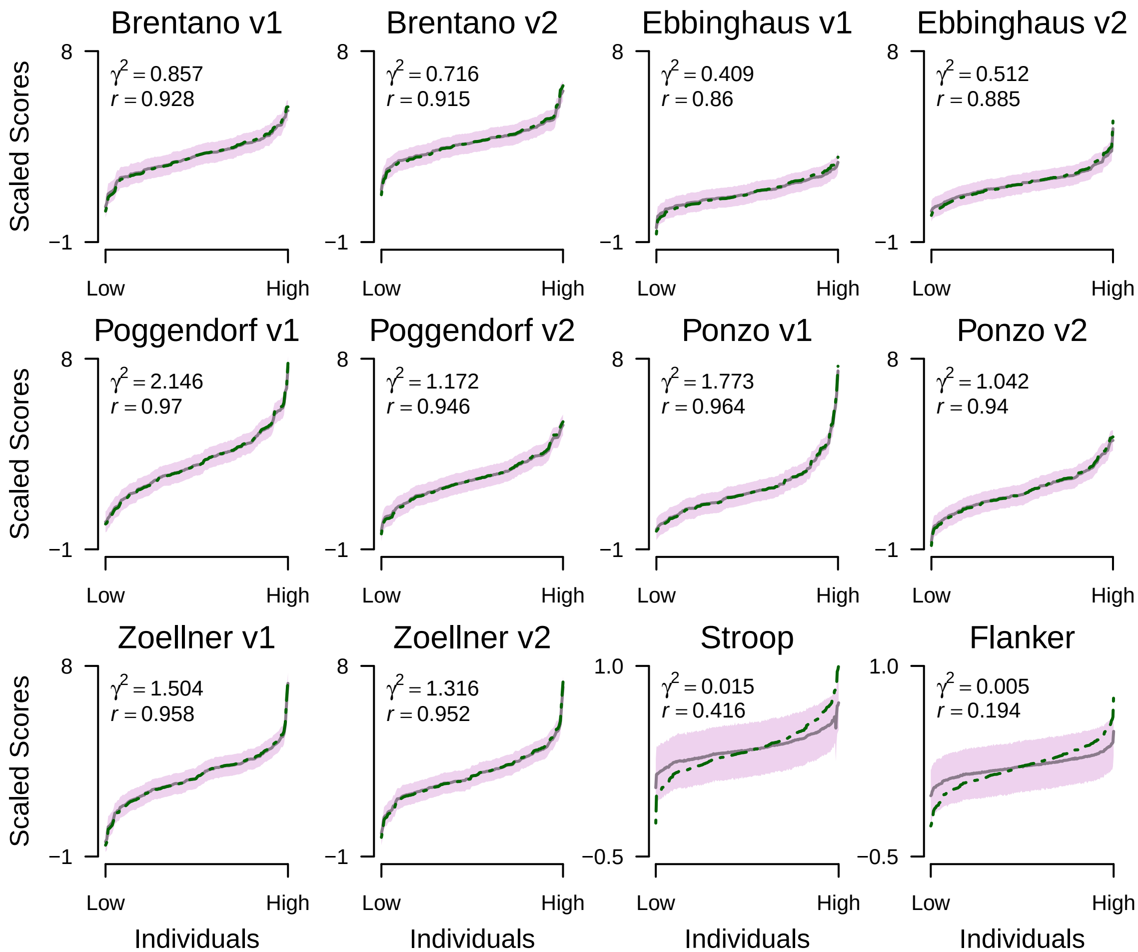

Figure 4

Task plots for the 10 illusion tasks and two cognitive control tasks. Observed effects are shown as dashed lines, while the solid lines represent model-based estimates. The shaded area indicates the 95% credible interval for the model-based effects. The task plots show individual effects ordered from smallest to largest. The illusion tasks demonstrate substantial variability in individual performance relative to noise, suggesting that studying individual differences is feasible even with a limited number of trials per task. In contrast, the two cognitive control tasks show only marginal individual differences when compared to trial noise.

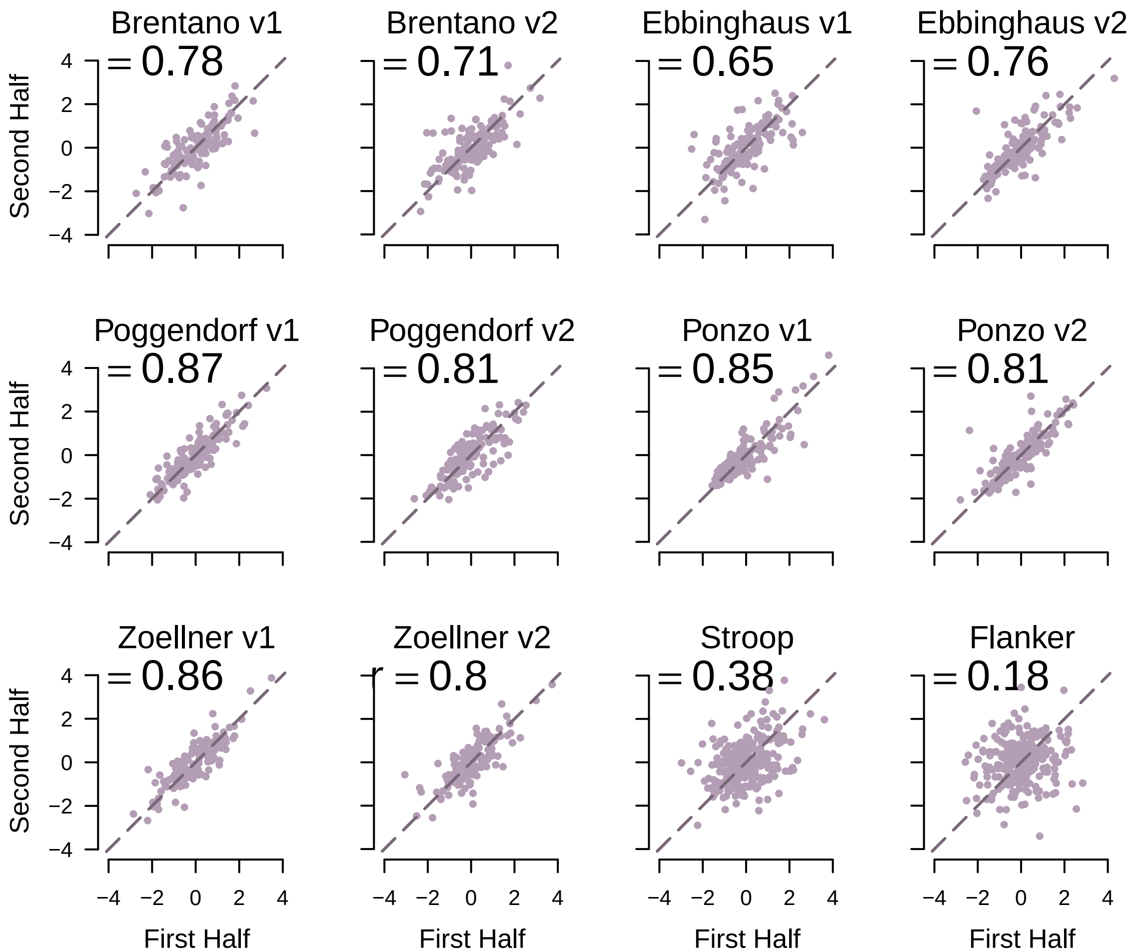

Figure 5

Split-half reliabilities serve as check. Even though the halves may differ systematically from learning and are comprised of 7 and 8 trials, respectively, the reliability is high. Split-half reliability for cognitive control is less even though there are over 10X as many trials, and this result again highlights the lack of trial noise in illusion tasks compared to cognitive control tasks.

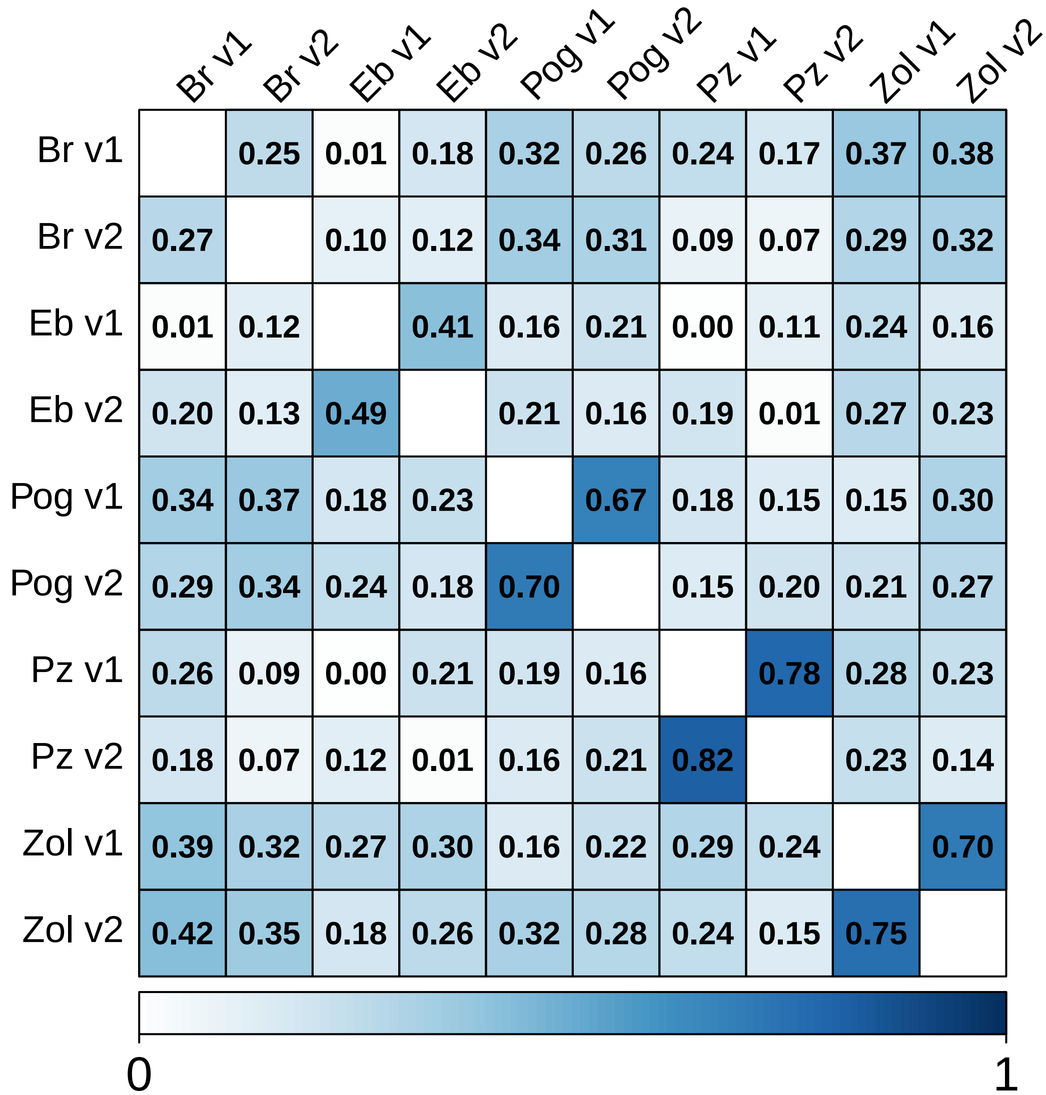

Figure 6

Correlations across tasks. The upper triangle shows sample correlations. The lower triangle is model-estimated correlations from the hierarchical Wishart model. Sample and model correlations are quite close reflecting a low degree of trial noise. The defining features are large correlations between versions for the Poggendorf, Ponzo, and Zöllner Illusions, but more marginal correlations for Brentano and Ebbinghaus Illusions.

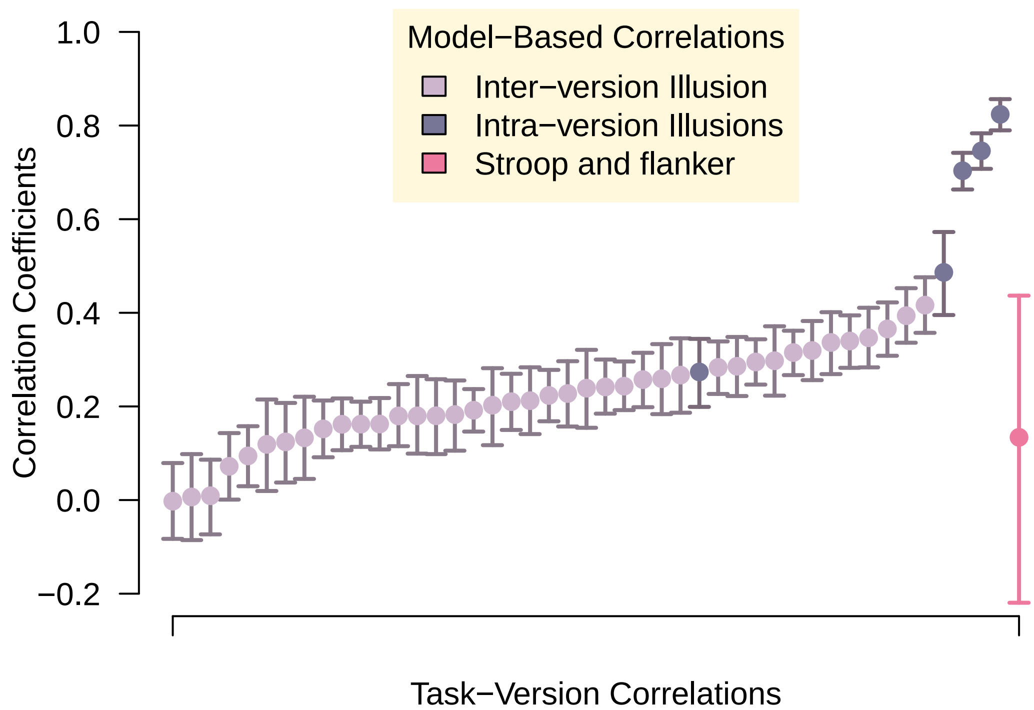

Figure 7

Uncertainties in correlations. Shown are model correlations (lower triangle in Figure 5) along with 95% credible intervals. The three version correlations are well separated; the cross-task correlations center around .22 in value.

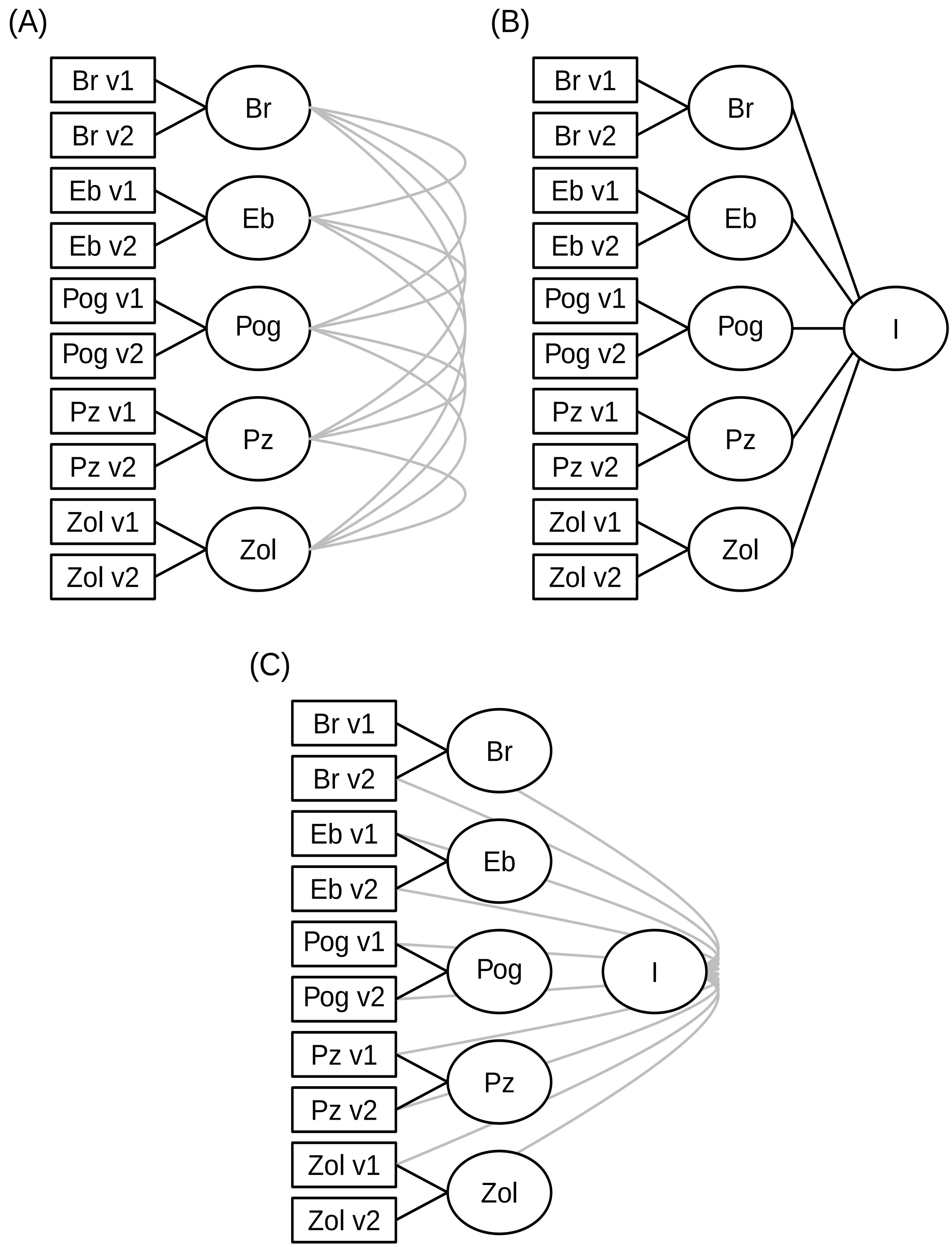

Figure 8

Three confirmatory factor models did not yield interpretable results.

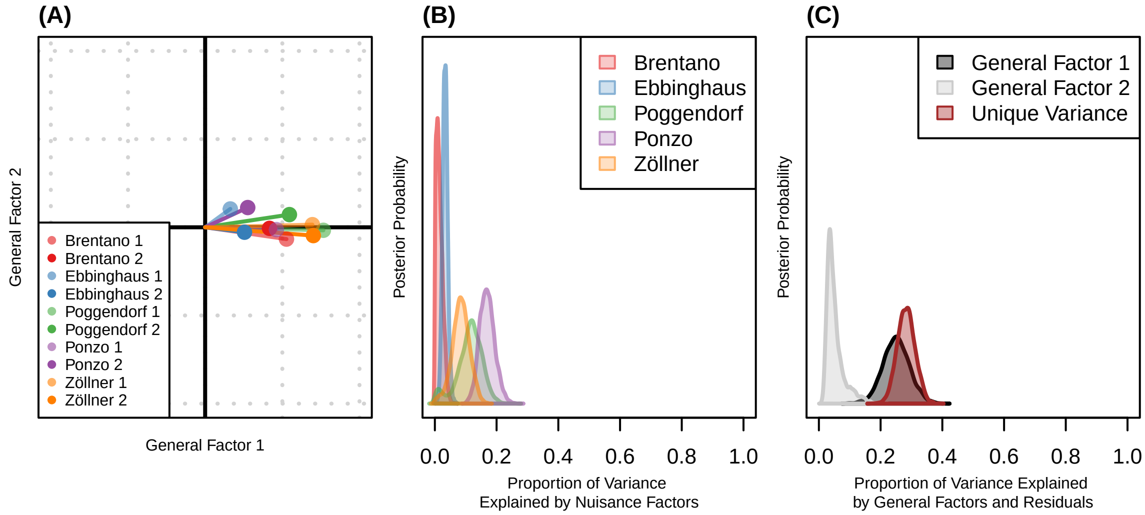

Figure 9

Summary of posterior estimates from the 7-factor hierarchical Bayesian model. (A) Task-wise projection onto the two general factors, where each arrow represents the average standardized loading of a task version on General Factor 1 (x-axis) and General Factor 2 (y-axis). (B) Posterior distributions of the proportion of total variance explained by each nuisance factor, which captures within-version idiosyncrasies for each illusion. (C) Posterior distributions of the proportion of total variance explained by the two general factors (black and gray), and the overall residual (unique) variance (red). All loadings and variance components are standardized to reflect proportions in the correlation metric.